

Learning goals: Strings in R

EXERCISES

This lab will use the data in from the file ‘worldfloras.txt’ and ‘HumanBones.txt’ files.

- (1) Given an integer, write a function to find if the integer is a palindrome.
- (2) Slice the string ‘seemerightnow’ to produce the following substrings: (a) ‘see’ (b) ‘me’ (c) ‘right’
- (3) Determine the fraction of G and C bases in the sequence “ATTGCGCATAGTCCGGG”.
- (4) Write a function to determine if a DNA nucleotide sequence is a palindrome in the sense that it is equal to its own complementary sequence read backward. For example, the sequence “TGGATCCA” is palindromic because its complement is “ACCTAGGT” which is same as the original sequence backward. The complementary base pairs are (A,T) and (G,C).
- (5) Write a code to search and print the largest word in this sentence: ”She sells hundreds of sea oysters on the sea shore.” By extension, print the second largest word in the same sentence. Extend this code to allow searching for words of equal length and gathering them into a list.
- (6) Load the data in ‘worldfloras.txt’ and do the following.
 - (a) Create subsets of countries within the same continent and store the data (ie. the allied columns) as different dataframes.
 - (b) Make a boxplot of the distribution of floral count within each continent and print the statistical summary. What are the mean and standard deviation values? Also calculate and comment on the skewness and kurtosis parameters (interpret them)
 - (c) Make a boxplot and histogram plot of the population distribution within each continent and print the statistical summary. Calculate and comment on the skewness and kurtosis parameters (interpret them). Does this have any relation with the floral count data?
- (7) Read in the data from ‘HumanBones.txt’ and group the data into categories “Chest”, “Spine”, “Skull”, “Ear Bones”, “Arms” and “Legs”. The number in the brackets indicates the number of bones in that type. Create a dataframe with 3 columns- category, name of the bone and number of bones.
- (8) Which category contains maximum number of bones? Create a frequency table and make a bar plot of each category.
- (9) Create a subset category of “Legs” bones and print the bone names longer than 5 letters.
- (10) List all the bones starting with “M” and substitute the lower-case “a” with upper-case “A”.
- (11) List all the bones ending with “e” and convert all the letters to lower-case.
- (12) List all the bones with two “o” s in their names.